

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Poster Presentation

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO3 –Poster Presentation	Assessment:	Assessment Tool 1– Poster Presentation
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	Asish.s.s	Reg. No.:	192424348
Section / Batch:	BTech AI&DS	Date:	20-08-2026

Code of Conduct

I Asish.s.s/192424348 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	



EXPLORING SKEWNESS AND OUTLIER DETECTION IN HEALTHCARE DATA

Data Interpretation using Descriptive Statistics and Exploratory Data Analysis

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1. INTRODUCTION

This study examines feature skewness and utilizes the Interquartile Range (IQR) method to detect statistical outliers in healthcare data. By identifying anomalies across vital clinical parameters, we ensure data quality and provide robust preprocessing for medical decision-making models.

PROBLEM STATEMENT

The presence of erroneous zero-values, heavy distributional skewness, and extreme physiological outliers in healthcare data compromises statistical integrity and degrades reliability of clinical diagnostic models.

OBJECTIVES

- Clean invalid zero measurements using median imputation.
- Analyze distributional shape and quantify feature skewness.
- Detect and count outliers using the robust Interquartile Range (IQR) method.
- Provide visual diagnostics via box, violin, and distribution plots.

2. DATASET DETAILS

Source: Pima Indians Diabetes Dataset

Dimensions: 768 Rows $\times$ 9 Columns

Key Predictors: Glucose, Blood Pressure, BMI, Insulin, Age, Skin Thickness, Pregnancies, Pedigree Function.

Target Variable: Outcome (0: Non-Diabetic [65.1%], 1: Diabetic [34.9%]).

Preprocessing Requirement: Zero-replacement and median imputation for physiologically impossible zero-values across 5 clinical attributes.

5. KEY FINDINGS

Highest Skewness & Outliers: Insulin (\$+3.385\$ skew, 346 outliers) shows extreme clinical variance.
Normal-like Feature: Blood Pressure (\$+0.145\$ skew, 14 outliers) exhibits near-normal symmetry.

3. METHODOLOGY / ANALYSIS PIPELINE

1. Data Collection

Collect healthcare data from Kaggle.

2. Data Cleaning

Check missing values and data quality.

3. Descriptive Statistics

Compute mean, median, std., min, max and quartiles.

4. Distribution Analysis

Plot histograms to study data distribution.

5. Skewness Calculation

Calculate skewness to identify asymmetry.

6. Outlier Detection

Detect outliers using the IQR method and box plots.

7. Interpretation & Insights

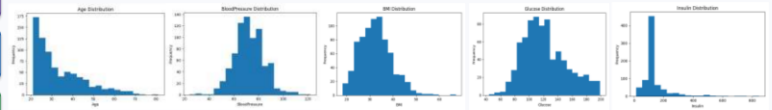
Summarize findings and observations.

OUTLIER DETECTION (IQR METHOD)

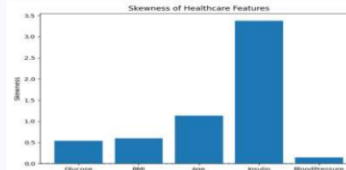
	Q1	Q3	IQR	Lower	Upper
Age	24.00	41.00	17.00	-1.50	66.50
BloodPressure	64.00	80.00	16.00	40.00	104.00
BMI	27.50	36.60	9.10	13.85	50.25
Glucose	99.75	140.25	40.50	39.00	201.00
Insulin	121.50	127.25	5.75	112.88	135.88

4. RESULTS & VISUAL ANALYSIS

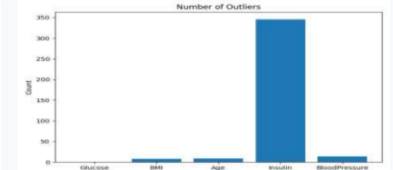
A. DISTRIBUTION ANALYSIS (HISTOGRAMS)



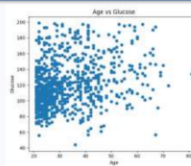
B. SKEWNESS ANALYSIS



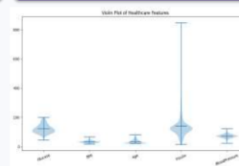
C. OUTLIER DETECTION



D. RELATIONSHIP ANALYSIS



E. DATA DISTRIBUTION



F. SUMMARY STATISTICS

Variable	Mean	Median	SD
Age	33.24	29.00	11.76
BloodPressure	72.39	72.00	12.10
BMI	32.46	32.30	6.88
Glucose	121.66	117.00	30.44
Insulin	140.67	125.00	86.38

6. CONCLUSION

The integration of median imputation and IQR-based outlier detection provides a reliable framework for managing skewed clinical distributions, ensuring enhanced data quality and model reliability in biomedical diagnostics.

7. FUTURE SCOPE

Future work involves integrating multivariate anomaly detection (Isolation Forest), applying log/Box-Cox transformations on skewed features, and benchmarking classification performance on predictive machine learning models.

Academic Poster • Fundamentals of Data Science • Data Interpretation / Exploratory Data Analysis